

Laboratorium
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Departemen Teknik Komputer
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Laporan Akhir

Praktikum Jaringan Komputer

Konfigurasi Vlan, OSPF Multivendor

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1 Langkah-Langkah Percobaan

1.1 Konfigurasi Cisco Switch

1. Masuk ke terminal Cisco Switch

2. Membuat Vlan

```
vlan 10
name Finance
exit
vlan 20
name HR
exit
vlan 30
name IT
exit
vlan 40
name Marketing
exit
```

3. Pastikan vlan sudah dibuat dan statusnya aktif

```
Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Gi1/2, Gi1/3
10	Finance	active	Gi0/2, Gi1/0
20	HR	active	Gi0/1
30	IT	active	Gi1/1
40	Marketing	active	Gi0/3
99	VLAN0099	active	
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

```
Switch#
```

4. Konfigurasi Access port untuk setiap vlan

```

Switch#conf t

interface gi0/2
description PC-FINANCE-VLAN10
switchport mode access
switchport access vlan 10
no shutdown
exit

interface gi0/1
description PC-HR-VLAN20
switchport mode access
switchport access vlan 20
no shutdown
exit

interface gi1/1
description PC-IT-VLAN30
switchport mode access
switchport access vlan 30
no shutdown
exit

interface gi0/3
description PC-MARKETING-VLAN40
switchport mode access
switchport access vlan 40
no shutdown
exit

```

5. Konfigurasi Trunk ke Cisco Router

```

Switch#conf t

interface gi0/0
switchport trunk encapsulation dot1q
description TRUNK-TO-CISCO-ROUTER
switchport mode trunk
switchport trunk allowed vlan 10,20,30,40
no shutdown
exit

```

1.2 Konfigurasi Cisco Router

1. Masuk ke Cisco Router
2. Konfigurasi Interface Trunk ke Switch

```
interface gig0/2
description TRUNK-TO-CISCO-SWITCH
no ip address
no shutdown
exit
```

3. Konfigurasi Subinterface VLAN 10

```
interface gig0/2.10
description GATEWAY-VLAN10-FINANCE
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
no shutdown
exit
```

4. Konfigurasi Subinterface VLAN 20

```
interface gig0/2.20
description GATEWAY-VLAN20-HR
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
no shutdown
exit
```

5. Konfigurasi Subinterface VLAN 30

```
interface gig0/2.30
description GATEWAY-VLAN30-IT
encapsulation dot1Q 30
ip address 192.168.30.1 255.255.255.0
no shutdown
exit
```

6. Konfigurasi Subinterface VLAN 40

```
interface gig0/2.40
description GATEWAY-VLAN40-MARKETING
encapsulation dot1Q 40
ip address 192.168.40.1 255.255.255.0
no shutdown
exit
```

7. Konfigurasi DHCP-Server untuk Vlan 10

```
ip dhcp excluded-address 192.168.10.1 192.168.10.10

ip dhcp pool VLAN10-FINANCE
network 192.168.10.0 255.255.255.0
default-router 192.168.10.1
dns-server 8.8.8.8
exit
```

8. Konfigurasi DHCP-Server untuk Vlan 20

```
ip dhcp excluded-address 192.168.20.1 192.168.20.10

ip dhcp pool VLAN20-HR
network 192.168.20.0 255.255.255.0
default-router 192.168.20.1
dns-server 8.8.8.8
exit
```

9. Konfigurasi IP Static pada VPCS untul vlan 30 dan 40

PC Vlan 30 :

```
ip 192.168.30.10/24 192.168.30.1
```

PC Vlan 40 :

```
ip 192.168.40.10/24 192.168.40.1
```

PC Vlan 10 dan 20 menggunakan dhcp:

```
ip dhcp
```

10. Verifikasi Cisco Router

```
Router#
Router#
Router#
Router#show ip int br
Router#show ip int brief
Interface      IP-Address      OK? Method Status        Protocol
GigabitEthernet0/0    unassigned      YES unset  administratively down  down
GigabitEthernet0/1    unassigned      YES unset  administratively down  down
GigabitEthernet0/2    unassigned      YES unset  up            up
GigabitEthernet0/2.10 192.168.10.1    YES manual  up            up
GigabitEthernet0/2.20 192.168.20.1    YES manual  up            up
GigabitEthernet0/2.30 192.168.30.1    YES manual  up            up
GigabitEthernet0/2.40 192.168.40.1    YES manual  up            up
GigabitEthernet0/3    unassigned      YES unset  administratively down  down
Router#
```

11. Pengujian dari VPCS

1.3 Konfigurasi Huawei Router dan IP Address Antarrouter

1. Konfigurasi Lan Huawei

```
system-view

interface Ethernet1/0/2
description LAN-HUAWEI
ip address 192.168.50.1 255.255.255.0
undo shutdown
quit

commit
```

2. Konfigurasi IP VPC Lan Huawei

```
ip 192.168.50.10/24 192.168.50.1
```

3. Konfigurasi IP link cisco <-> Huawei link 1

Cisco Router

```
enable
configure terminal

interface gi0/3
description LINK1-TO-HUAWEI
ip address 10.10.10.1 255.255.255.252
no shutdown
exit
```

Huawei Router

```
system-view

interface Ethernet1/0/0
description LINK1-TO-CISCO
ip address 10.10.10.2 255.255.255.252
undo shutdown
quit

commit
```

4. Konfigurasi IP Link Cisco Huawei Link 2

Cisco Router

```
interface gi0/0
description LINK2-TO-HUAWEI
ip address 10.10.11.1 255.255.255.252
no shutdown
exit
```

Huawei Router

```
system-view

interface Ethernet1/0/4
description LINK2-TO-CISCO
ip address 10.10.11.2 255.255.255.252
undo shutdown
quit

commit
```

5. Konfigurasi IP link Huawei <-> Mikrotik

```

Huawei Router

system-view

interface Ethernet1/0/3
description LINK-TO-MIKROTIK
ip address 10.10.20.1 255.255.255.252
undo shutdown
quit

commit

Mikrotik

/ip address add address=10.10.20.2/30 interface=ether1 comment=TO-HUAWEI

```

6. Konfigurasi IP link Cisco <-> Mikrotik

```

Cisco Router

interface gi0/1
description LINK-TO-MIKROTIK
ip address 10.10.30.1 255.255.255.252
no shutdown
exit

Mikrotik

/ip address add address=10.10.30.2/30 interface=ether2 comment=TO-CISCO

```

7. Verifikasi Interface

8. Verifikasi ping Antar Router

1.4 Konfigurasi OSPF Multivendor dan OSPF Cost

1. Konsep routing ospf pada topologi
2. Network yang diiklankan ke ospf
3. Konfigurasi ospf pada Cisco Router

```

router ospf 1
router-id 1.1.1.1
network 10.10.10.0 0.0.0.3 area 0
network 10.10.11.0 0.0.0.3 area 0
network 10.10.30.0 0.0.0.3 area 0
network 192.168.10.0 0.0.0.255 area 0
network 192.168.20.0 0.0.0.255 area 0
network 192.168.30.0 0.0.0.255 area 0
network 192.168.40.0 0.0.0.255 area 0
exit

```

4. Konfigurasi OSPF pada Huawei Router

Konfigurasi OSPF:

```
ospf 1 router-id 2.2.2.2
area 0.0.0.0
network 10.10.10.0 0.0.0.3
network 10.10.11.0 0.0.0.3
network 10.10.20.0 0.0.0.3
network 192.168.50.0 0.0.0.255
quit
quit
```

Atur OSPF cost pada interface huawei.

```
interface Ethernet1/0/0
description LINK1-TO-CISCO
ospf cost 10
quit

interface Ethernet1/0/4
description LINK2-TO-CISCO
ospf cost 10
quit

interface Ethernet1/0/3
description LINK-TO-MIKROTIK
ospf cost 100
quit

commit
```

5. Konfigurasi OSPF pada MikroTik
6. Verifikasi OSPF Neighbor
7. Verifikasi Routing Table
8. Verifikasi Koneksi end-to-end

1.5 Pengujian Failover OSPF

1. Kondisi normal: Dua link cisco-huawei aktif

```
VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.10.1    2.664 ms  1.766 ms  1.699 ms
 2  10.10.10.2     5.802 ms  2.401 ms  2.825 ms
 3  *192.168.50.10 17.171 ms (ICMP type:3, code:3, Destination port unreachable)
VPCS> █
```

2. Kondisi 1 link cisco-huawei mati

```
VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.10.1    6.264 ms  2.149 ms  2.647 ms
 2  10.10.11.2     3.815 ms  2.709 ms  1.908 ms
 3  *192.168.50.10 14.584 ms (ICMP type:3, code:3, Destination port unreachable)
VPCS> █
```


3. Kondisi dua link cisco-huawei mati

```
VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.10.1      2.501 ms  2.305 ms  2.885 ms
 2  10.10.30.2       2.714 ms  1.933 ms  2.174 ms
 3  10.10.20.1       4.843 ms  2.507 ms  2.775 ms
 4  *192.168.50.10   3.970 ms (ICMP type:3, code:3, Destination port unreachable)
VPCS>
```

4. Mengaktifkan kembali dua link cisco huawei

```
VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.10.1      5.783 ms  8.876 ms  4.795 ms
 2  10.10.11.2       9.134 ms  2.152 ms  2.584 ms
 3  *192.168.50.10   10.945 ms (ICMP type:3, code:3, Destination port unreachable)
VPCS>
```

2 Analisis Hasil Percobaan

Pada percobaan pertama yaitu konfigurasi vlan dan access port berdasarkan hasil praktikum, konfigurasi VLAN pada Cisco Switc berhasil sesuai dengan teori. Hasil verifikasi dengan perintah 'show vlan brief' menunjukkan bahwa semua VLAN aktif dan siap digunakan. Konfigurasi access port untuk setiap VLAN juga telah dikonfigurasi dengan tepat sesuai dengan topologi faktor keberhasilan ini adalah ketelitian dalam memasukkan konfigurasi, tidak terdapat kendala pada saat percobaan ini tetapi harus sangat fokus agar tidak salah mengetik konfigurasi. Pada percobaan selanjutnya mengenai trunk port, Konfigurasi trunk port antara Cisco Switch dan Cisco Router berhasil menerapkan enkapsulasi 802.1Q dengan membawa empat vlan secara bersamaan melalui satu koneksi fisik. Hal ini sesuai dengan teori bahwa trunk port dapat mengoptimalkan penggunaan bandwidth dengan mengirim traffic dari multiple VLAN melalui satu link. mplementasi ROAS (Router-on-a-Stick) pada Cisco Router dengan membuat subinterface untuk setiap VLAN menunjukkan hasil yang sesuai dengan teori. Setiap subinterface (gig0/2.10, gig0/2.20, gig0/2.30, gig0/2.40) dikonfigurasi dengan gateway masing-masing VLAN, dan hasil verifikasi menunjukkan semua interface aktif dengan IP address yang benar.

Pada percobaan kedua mengenai Konfigurasi DHCP Server pada Cisco Router untuk VLAN 10 dan VLAN 20 berhasil dengan hasil yang sesuai dengan teori. PC pada VLAN 10 dan 20 berhasil mendapatkan IP address secara otomatis melalui mekanisme DHCP, dengan IP dalam range yang telah dikonfigurasi (192.168.10.0/24 dan 192.168.20.0/24). Sementara itu, PC pada VLAN 30 dan 40 di-konfigurasi dengan static IP, yang juga berfungsi dengan baik. Pengujian ping dari masing-masing PC ke gateway menunjukkan konektivitas yang sempurna membuktikan bahwa routing antar VLAN sudah berfungsi.

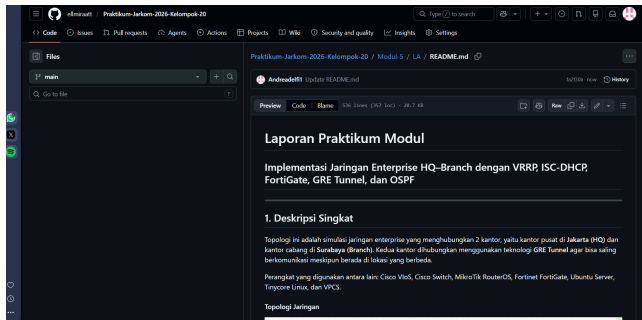
Pada percobaan ketiga Konfigurasi alamat IP pada link antar router dengan menggunakan subnet /30 yang sesuai untuk point to point link. menggunakan /30 karena hanya membutuhkan 4 alamat ip. semua interface aktif dan dapat saling ping dengan sukses. MikroTik menunjukkan konfigurasi IP yang konsisten dengan tabel addressing yang diberikan.

Pada percobaan keempat Implementasi OSPF pada tiga perangkat berbeda menunjukkan protokol routing dinamis dapat bekerja dalam lingkungan multivendor sesuai dengan standar IEEE dan RFC OSPF. Hasil verifikasi menunjukkan bahwa Cisco Router membentuk neighbor dengan Huawei Router melalui dua link utama dan satu link backup ke MikroTik, sesuai dengan topologi. Cost yang dikonfigurasi berhasil mempengaruhi path selection sesuai dengan teori algoritma djikstra shortest path.

Pada percobaan kelima Pengujian failover OSPF menunjukkan bahwa mekanisme convergence dinamis bekerja sesuai teori. Ketika satu link Cisco-Huawei dimatikan outing table tetap memiliki path

menuju LAN Huawei melalui link cadangan yang tersisa (gi0/0). Traffic dari PC VLAN masih dapat mencapai LAN Huawei tanpa gangguan, menunjukkan redundancy yang efektif. Ketika dua link Cisco-Huawei dimatikan, traffic secara otomatis beralih ke jalur backup melalui MikroTik dengan cost total 200. race dari PC VLAN menunjukkan perubahan hop count sesuai dengan kondisi link, membuktikan bahwa OSPF melakukan reconvergence dengan cepat. Ketika kedua link utama diaktifkan kembali, traffic kembali beralih ke jalur utama Cisco-Huawei, menunjukkan bahwa OSPF memilih jalur terbaik secara realtime.

3 Hasil Tugas Modul



4 Kesimpulan

Praktikum jaringan komputer multivendor dengan VLAN dan OSPF telah berhasil dilakukan dengan hasil yang sesuai dengan teori. Konfigurasi VLAN dan access/trunk port telah membuktikan bahwa network segmentation dapat meningkatkan security dan manageability. Implementasi ROAS dan DHCP server menunjukkan bahwa single gateway dapat melayani multiple VLAN secara efisien. Konfigurasi IP pada link antar router dengan benar memastikan dasar yang solid untuk routing dinamis. Implementasi OSPF pada environment multivendor telah membuktikan bahwa standar industri seperti RFC OSPF memungkinkan interoperabilitas antar perangkat dari vendor berbeda. Terakhir, pengujian failover menunjukkan bahwa OSPF memberikan keandalan tinggi dengan automatic failover dan rapid convergence yang esensial untuk production network. Hasil praktikum ini memvalidasi teori-teori yang dipelajari dan menunjukkan aplikasinya dalam dunia asli.

5 Lampiran

5.1 Dokumentasi saat praktikum

```
Switch#show vlan brief
Switch#show vlan brief
VLAN Name                Status Ports
-----
1    default              active  Gi0/0, Gi0/1, Gi0/2, Gi0/3
10   FINANCE               active  Gi1/0, Gi1/1, Gi1/2, Gi1/3
20   HR                    active
30   IT                     active
40   MARKETING              active
1002 fddi-default          act/unsup
1003 token-ring-default    act/unsup
1004 fddinet-default       act/unsup
1005 trnet-default         act/unsup
Switch#

Router#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  unassigned      YES unset  administratively down down
GigabitEthernet0/1  unassigned      YES unset  administratively down down
GigabitEthernet0/2  unassigned      YES unset  up          up
GigabitEthernet0/2.10  192.168.10.1  YES manual up          up
GigabitEthernet0/2.20  192.168.20.1  YES manual up          up
GigabitEthernet0/2.30  192.168.30.1  YES manual up          up
GigabitEthernet0/2.40  192.168.40.1  YES manual up          up
GigabitEthernet0/3  unassigned      YES unset  administratively down down
Router#

VPCS> show ip
NAME          : VPCS[1]
IP/MASK       : 192.168.10.11/24
GATEWAY       : 192.168.10.1
DNS           : 8.8.8.8
DHCP SERVER   : 192.168.10.1
DHCP LEASE    : 86231, 86400/43200/75600
MAC           : 00:50:79:66:68:84
LPORT        : 20000
RHOST:PORT    : 127.0.0.1:30000
MTU           : 1500

VPCS>

VPCS> show ip
NAME          : VPCS[1]
IP/MASK       : 192.168.20.11/24
GATEWAY       : 192.168.20.1
DNS           : 8.8.8.8
DHCP SERVER   : 192.168.20.1
DHCP LEASE    : 86172, 86400/43200/75600
MAC           : 00:50:79:66:68:83
LPORT        : 20000
RHOST:PORT    : 127.0.0.1:30000
MTU           : 1500

VPCS>

VPCS> ping 192.168.30.1
84 bytes from 192.168.30.1 icmp_seq=1 ttl=255 time=3.134 ms
84 bytes from 192.168.30.1 icmp_seq=2 ttl=255 time=4.138 ms
84 bytes from 192.168.30.1 icmp_seq=3 ttl=255 time=4.657 ms
84 bytes from 192.168.30.1 icmp_seq=4 ttl=255 time=2.955 ms
84 bytes from 192.168.30.1 icmp_seq=5 ttl=255 time=5.074 ms
^C

VPCS>

VPCS> ping 192.168.20.11
84 bytes from 192.168.20.11 icmp_seq=1 ttl=63 time=5.493 ms
84 bytes from 192.168.20.11 icmp_seq=2 ttl=63 time=6.778 ms
84 bytes from 192.168.20.11 icmp_seq=3 ttl=63 time=5.525 ms
84 bytes from 192.168.20.11 icmp_seq=4 ttl=63 time=6.121 ms
84 bytes from 192.168.20.11 icmp_seq=5 ttl=63 time=5.673 ms

VPCS> ping 192.168.30.10
84 bytes from 192.168.30.10 icmp_seq=1 ttl=63 time=5.860 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=63 time=5.465 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=63 time=4.598 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=63 time=6.202 ms
84 bytes from 192.168.30.10 icmp_seq=5 ttl=63 time=5.523 ms
^[[A
VPCS> ping 192.168.40.10
84 bytes from 192.168.40.10 icmp_seq=1 ttl=63 time=5.181 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=63 time=5.263 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=63 time=4.893 ms
84 bytes from 192.168.40.10 icmp_seq=4 ttl=63 time=5.073 ms
84 bytes from 192.168.40.10 icmp_seq=5 ttl=63 time=5.731 ms

VPCS>

VPCS> ping 192.168.50.1
84 bytes from 192.168.50.1 icmp_seq=1 ttl=255 time=9.769 ms
84 bytes from 192.168.50.1 icmp_seq=2 ttl=255 time=3.222 ms
84 bytes from 192.168.50.1 icmp_seq=3 ttl=255 time=3.317 ms
84 bytes from 192.168.50.1 icmp_seq=4 ttl=255 time=3.757 ms
84 bytes from 192.168.50.1 icmp_seq=5 ttl=255 time=3.085 ms

VPCS>

Switch#show interfaces trunk
Port      Mode          Encapsulation  Status        Native vlan
Gi0/0     on            802.1q         trunking      1

Port      Vlans allowed on trunk
Gi0/0     10,20,30,40

Port      Vlans allowed and active in management domain
Gi0/0     10,20,30,40

Port      Vlans in spanning tree forwarding state and not pruned
Gi0/0     10,20,30,40
Switch#

Router#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration        Type
                Hardware address/
                User name
192.168.10.11   0100.5079.6668.84  Jun 06 2026 02:37 AM    Automatic
192.168.20.11   0100.5079.6668.83  Jun 06 2026 02:37 AM    Automatic
Router#

VPCS> ping 192.168.10.1
84 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=3.620 ms
84 bytes from 192.168.10.1 icmp_seq=2 ttl=255 time=3.333 ms
84 bytes from 192.168.10.1 icmp_seq=3 ttl=255 time=3.840 ms
84 bytes from 192.168.10.1 icmp_seq=4 ttl=255 time=3.083 ms
^C

VPCS>

VPCS> ping 192.168.20.1
84 bytes from 192.168.20.1 icmp_seq=1 ttl=255 time=3.074 ms
84 bytes from 192.168.20.1 icmp_seq=2 ttl=255 time=3.001 ms
84 bytes from 192.168.20.1 icmp_seq=3 ttl=255 time=3.445 ms
84 bytes from 192.168.20.1 icmp_seq=4 ttl=255 time=5.182 ms
^C
VPCS>

VPCS> ping 192.168.40.1
84 bytes from 192.168.40.1 icmp_seq=1 ttl=255 time=3.336 ms
84 bytes from 192.168.40.1 icmp_seq=2 ttl=255 time=3.413 ms
84 bytes from 192.168.40.1 icmp_seq=3 ttl=255 time=4.277 ms
84 bytes from 192.168.40.1 icmp_seq=4 ttl=255 time=3.569 ms
84 bytes from 192.168.40.1 icmp_seq=5 ttl=255 time=3.007 ms

VPCS>

VPCS> ip 192.168.50.10/24 192.168.50.1
Checking for duplicate address...
PC1 : 192.168.50.10 255.255.255.0 gateway 192.168.50.1

VPCS> show ip
NAME          : VPCS[1]
IP/MASK       : 192.168.50.10/24
GATEWAY       : 192.168.50.1
DNS           :
MAC           : 00:50:79:66:68:85
LPORT        : 20000
RHOST:PORT    : 127.0.0.1:30000
MTU           : 1500

VPCS>

Router#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  10.10.11.1      YES manual up          up
GigabitEthernet0/1  10.10.30.1      YES manual up          up
GigabitEthernet0/2  unassigned      YES unset  up          up
GigabitEthernet0/2.10  192.168.10.1  YES manual up          up
GigabitEthernet0/2.20  192.168.20.1  YES manual up          up
GigabitEthernet0/2.30  192.168.30.1  YES manual up          up
GigabitEthernet0/2.40  192.168.40.1  YES manual up          up
GigabitEthernet0/3  10.10.10.1      YES manual up          up
Router#
```

```

[~HUAWEI]display ip interface brief
down: administratively down
down: FIB overload down
down: standby
(l): loopback
(s): spoofing
(d): Dampening Suppressed
(E): E-Trunk down
the number of interface that is UP in Physical is 23
the number of interface that is DOWN in Physical is 0
the number of interface that is UP in Protocol is 16
the number of interface that is DOWN in Protocol is 7

Interface                IP Address/Mask      Physical  Protocol VPN
Ethernet1/0/0            10.10.10.2/30        up        up      --
Ethernet1/0/0.4094       128.1.138.137/16     up        up      l3vpn
Ethernet1/0/1            128.1.138.137/16     up        down    --
Ethernet1/0/1.4094       128.1.138.137/16     up        up      l3vpn
Ethernet1/0/2            192.168.50.1/24      up        up      --
Ethernet1/0/2.4094       128.1.138.137/16     up        up      l3vpn
Ethernet1/0/3            10.10.20.1/30        up        up      --
Ethernet1/0/3.4094       128.1.138.137/16     up        up      l3vpn
Ethernet1/0/4            10.10.11.2/30        up        up      --
Ethernet1/0/4.4094       128.1.138.137/16     up        up      l3vpn
---- More ----

```

```

[admin@MikroTik] > ping 10.10.20.1
SEQ HOST                SIZE TTL TIME STATUS
0 10.10.20.1            56 255 6ms  success
1 10.10.20.1            56 255 1ms  success
2 10.10.20.1            56 255 1ms  success
3 10.10.20.1            56 255 1ms  success
4 10.10.20.1            56 255 1ms  success
5 10.10.20.1            56 255 1ms  success
6 10.10.20.1            56 255 1ms  success
7 10.10.20.1            56 255 1ms  success
sent=8 received=8 packet-loss=0% min-rtt=1ms avg-rtt=1ms max-rtt=6ms

[admin@MikroTik] > ping 10.10.30.1
SEQ HOST                SIZE TTL TIME STATUS
0 10.10.30.1            56 255 5ms  success
1 10.10.30.1            56 255 1ms  success
2 10.10.30.1            56 255 1ms  success
3 10.10.30.1            56 255 1ms  success
4 10.10.30.1            56 255 1ms  success
5 10.10.30.1            56 255 1ms  success
6 10.10.30.1            56 255 1ms  success
sent=7 received=7 packet-loss=0% min-rtt=1ms avg-rtt=1ms max-rtt=5ms

[admin@MikroTik] >

```

```

Router#ping 10.10.11.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.11.2, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

```

```

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.10.10.0 0.0.0.3 area 0
    10.10.11.0 0.0.0.3 area 0
    10.10.30.0 0.0.0.3 area 0
    192.168.10.0 0.0.0.255 area 0
    192.168.20.0 0.0.0.255 area 0
    192.168.30.0 0.0.0.255 area 0
    192.168.40.0 0.0.0.255 area 0
  Routing Information Sources:
    Gateway         Distance         Last Update
  Distance: (default is 110)

Router#

```

```

[~HUAWEI]display ip routing-table protocol ospf
Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route
-----
public_Routing Table : OSPF
  Destinations : 9          Routes : 14

OSPF routing table status : <Active>
  Destinations : 5          Routes : 10

Destination/Mask    Proto Pre Cost      Flags NextHop        Interface
-----
10.10.30.0/30       OSPF  10  110          D  10.10.11.1 Ethernet1/0/4
OSPF                10  110          D  10.10.10.1 Ethernet1/0/4
192.168.10.0/24      OSPF  10  11          D  10.10.10.1 Ethernet1/0/0
OSPF                10  11          D  10.10.11.1 Ethernet1/0/4
192.168.20.0/24      OSPF  10  11          D  10.10.10.1 Ethernet1/0/0
OSPF                10  11          D  10.10.11.1 Ethernet1/0/4
192.168.30.0/24      OSPF  10  11          D  10.10.10.1 Ethernet1/0/0
OSPF                10  11          D  10.10.11.1 Ethernet1/0/4
192.168.40.0/24      OSPF  10  11          D  10.10.11.1 Ethernet1/0/4
OSPF                10  11          D  10.10.10.1 Ethernet1/0/0

OSPF routing table status : <Inactive>
  Destinations : 4          Routes : 4

Destination/Mask    Proto Pre Cost      Flags NextHop        Interface
-----
10.10.10.0/30       OSPF  10  10          D  10.10.10.2 Ethernet1/0/0
10.10.11.0/30       OSPF  10  10          D  10.10.11.2 Ethernet1/0/4
10.10.20.0/30       OSPF  10  100         D  10.10.20.1 Ethernet1/0/3
192.168.50.0/24     OSPF  10  1          D  192.168.50.1 Ethernet1/0/2

[~HUAWEI]

```

```

[admin@MikroTik] > /routing ospf neighbor print
0 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1
  dr-address=10.10.20.1 backup-dr-address=0.0.0.0 state="ExStart" state-changes=3
  ls-retransmits=0 ls-requests=0 db-summaries=0

1 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1
  dr-address=10.10.30.1 backup-dr-address=0.0.0.0 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=6s

[admin@MikroTik] >

```

```

[admin@MikroTik] > ip address print
Flags: X - disabled, I - invalid, D - dynamic
# ADDRESS NETWORK INTERFACE
0 ;; TO-HUAWEI 10.10.20.2/30 10.10.20.0 ether1
1 ;; TO-CISCO 10.10.30.2/30 10.10.30.0 ether2
[admin@MikroTik] >

```

```

Router#ping 10.10.30.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.30.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/3 ms
Router#

```

```

Router#ping 10.10.10.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

```

```

(M) Indicates MADJ neighbor

OSPF Process 1 with Router ID 2.2.2.2
  Neighbors

Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1 Address: 10.10.10.1
State: Full Mode:Nbr is Slave Priority: 1
DR: 10.10.10.2 BDR: 10.10.10.1 MTU: 1500
Dead timer due in 32 sec
Retrans timer interval: 5
Neighbor is up for 00h00m34s
Neighbor Up Time : 2026-06-05 03:25:13
Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
Router ID: 1.1.1.1 Address: 10.10.11.1
State: Full Mode:Nbr is Slave Priority: 1
DR: 10.10.11.2 BDR: 10.10.11.1 MTU: 1500
Dead timer due in 34 sec
Retrans timer interval: 5
Neighbor is up for 00h00m39s
Neighbor Up Time : 2026-06-05 03:25:08
Authentication Sequence: [ 0 ]

[~HUAWEI]
[~HUAWEI]

```

```

[~HUAWEI]display ospf routing

OSPF Process 1 with Router ID 2.2.2.2
  Routing Tables

Routing for Network
Destination Cost Type NextHop AdvRouter Area
-----
10.10.10.0/30 10 Direct 10.10.10.2 2.2.2.2 0.0.0.0
10.10.11.0/30 10 Direct 10.10.11.2 2.2.2.2 0.0.0.0
10.10.20.0/30 100 Direct 10.10.20.1 2.2.2.2 0.0.0.0
10.10.30.0/30 110 Stub 10.10.11.1 1.1.1.1 0.0.0.0
10.10.30.0/30 110 Stub 10.10.10.1 1.1.1.1 0.0.0.0
192.168.10.0/24 11 Stub 10.10.11.1 1.1.1.1 0.0.0.0
192.168.10.0/24 11 Stub 10.10.10.1 1.1.1.1 0.0.0.0
192.168.20.0/24 11 Stub 10.10.11.1 1.1.1.1 0.0.0.0
192.168.20.0/24 11 Stub 10.10.10.1 1.1.1.1 0.0.0.0
192.168.30.0/24 11 Stub 10.10.11.1 1.1.1.1 0.0.0.0
192.168.30.0/24 11 Stub 10.10.10.1 1.1.1.1 0.0.0.0
192.168.40.0/24 11 Stub 10.10.11.1 1.1.1.1 0.0.0.0
192.168.40.0/24 11 Stub 10.10.10.1 1.1.1.1 0.0.0.0
192.168.50.0/24 1 Direct 192.168.50.1 2.2.2.2 0.0.0.0

Total Nets: 9
Intra Area: 9 Inter Area: 0 ASE: 0 NSSA: 0
---- More ----

```

```

[admin@MikroTik] > ip route print where ospf
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS PREF-SRC GATEWAY DISTANCE
0 Ado 10.10.10.0/30 10.10.30.1 110
1 Ado 10.10.11.0/30 10.10.20.1 110
2 Ado 192.168.10.0/24 10.10.20.1 110
3 Ado 192.168.20.0/24 10.10.30.1 110
4 Ado 192.168.30.0/24 10.10.30.1 110
5 Ado 192.168.40.0/24 10.10.30.1 110
6 Ado 192.168.50.0/24 10.10.20.1 110

[admin@MikroTik] >

```

```
Router#show ip ospf neighbor

Neighbor ID    Pri   State           Dead Time   Address        Interface
3.3.3.3        1     FULL/BDR        00:00:37   10.10.30.2    GigabitEthernet0/1
2.2.2.2        1     FULL/DR         00:00:32   10.10.11.2    GigabitEthernet0/0
2.2.2.2        1     FULL/DR         00:00:37   10.10.10.2    GigabitEthernet0/3
Router#
```

```
[admin@MikroTik] > routing ospf neighbor print
0 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1
  dr-address=10.10.20.1 backup-dr-address=10.10.20.2 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=1m54s
1 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1
  dr-address=10.10.30.1 backup-dr-address=10.10.30.2 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=2m3s
[admin@MikroTik] >
```

```
[~HUAWEI]display ip routing-table protocol ospf
Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route
.....
public_Routing Table : OSPF
Destinations : 9      Routes : 14

OSPF routing table status : <Active>
Destinations : 5      Routes : 10

Destination/Mask    Proto  Pre  Cost      Flags NextHop        Interface
10.10.30.0/30       OSPF   10   110        D  10.10.11.1    Ethernet1/0/4
192.168.10.0/24     OSPF   10   110        D  10.10.11.1    Ethernet1/0/0
192.168.10.0/24     OSPF   10   11          D  10.10.11.1    Ethernet1/0/4
192.168.20.0/24     OSPF   10   11          D  10.10.11.1    Ethernet1/0/4
192.168.30.0/24     OSPF   10   11          D  10.10.11.1    Ethernet1/0/4
192.168.40.0/24     OSPF   10   11          D  10.10.11.1    Ethernet1/0/4
192.168.40.0/24     OSPF   10   11          D  10.10.11.1    Ethernet1/0/4

OSPF routing table status : <Inactive>
Destinations : 4      Routes : 4

Destination/Mask    Proto  Pre  Cost      Flags NextHop        Interface
10.10.10.0/30       OSPF   10   10          D  10.10.10.2    Ethernet1/0/0
10.10.11.0/30       OSPF   10   10          D  10.10.11.2    Ethernet1/0/4
10.10.20.0/30       OSPF   10   100         D  10.10.20.1    Ethernet1/0/3
192.168.50.0/24     OSPF   10   1          D  192.168.50.1  Ethernet1/0/2
[~HUAWEI]
```

```
VPCS> ping 192.168.50.10
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=12.786 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=11.120 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=6.798 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=6.112 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=7.819 ms
VPCS>
```

```
VPCS> ping 192.168.50.10
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=5.533 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=6.046 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=5.577 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=5.945 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=6.033 ms
VPCS>
```

```
VPCS> ping 192.168.10.11
84 bytes from 192.168.10.11 icmp_seq=1 ttl=62 time=7.906 ms
84 bytes from 192.168.10.11 icmp_seq=2 ttl=62 time=5.975 ms
84 bytes from 192.168.10.11 icmp_seq=3 ttl=62 time=8.083 ms
84 bytes from 192.168.10.11 icmp_seq=4 ttl=62 time=5.869 ms
84 bytes from 192.168.10.11 icmp_seq=5 ttl=62 time=9.098 ms
VPCS>
```

```
VPCS> ping 192.168.20.11
84 bytes from 192.168.20.11 icmp_seq=1 ttl=62 time=7.081 ms
84 bytes from 192.168.20.11 icmp_seq=2 ttl=62 time=10.277 ms
84 bytes from 192.168.20.11 icmp_seq=3 ttl=62 time=8.245 ms
84 bytes from 192.168.20.11 icmp_seq=4 ttl=62 time=6.958 ms
84 bytes from 192.168.20.11 icmp_seq=5 ttl=62 time=8.759 ms
VPCS>
```

```
VPCS> ping 192.168.30.10
84 bytes from 192.168.30.10 icmp_seq=1 ttl=62 time=6.784 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=62 time=6.938 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=62 time=6.912 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=62 time=9.874 ms
84 bytes from 192.168.30.10 icmp_seq=5 ttl=62 time=7.791 ms
VPCS>
```

```
VPCS> ping 192.168.40.10
84 bytes from 192.168.40.10 icmp_seq=1 ttl=62 time=6.412 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=62 time=7.612 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=62 time=8.503 ms
84 bytes from 192.168.40.10 icmp_seq=4 ttl=62 time=7.285 ms
84 bytes from 192.168.40.10 icmp_seq=5 ttl=62 time=6.246 ms
VPCS>
```

```
VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.40.1 4.012 ms 5.762 ms 4.789 ms
 2 10.10.11.2 9.748 ms 5.180 ms 3.660 ms
 3 *192.168.50.10 6.658 ms (ICMP type:3, code:3, Destination port unreachable)
```

```
-HUAWEI]display ospf peer

M) Indicates MADJ neighbor

OSPF Process 1 with Router ID 2.2.2.2
Neighbors

Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1 Address: 10.10.10.1
State: Full Mode:Nbr is Slave Priority: 1
DR: 10.10.10.2 BDR: 10.10.10.1 MTU: 1500
Dead timer due in 35 sec
Retrans timer interval: 5
Neighbor is up for 00h04m14s
Neighbor Up Time : 2026-06-05 03:25:13
Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.20.1 (Eth1/0/3)'s neighbors
Router ID: 3.3.3.3 Address: 10.10.20.2
State: Full Mode:Nbr is Master Priority: 1
DR: 10.10.20.1 BDR: 10.10.20.2 MTU: 1500
Dead timer due in 32 sec
Retrans timer interval: 5
Neighbor is up for 00h01m05s
Neighbor Up Time : 2026-06-05 03:28:22
Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
---- More ----
```

```
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
Known via "ospf 1", distance 110, metric 11, type intra area
Last update from 10.10.11.2 on GigabitEthernet0/0, 00:05:09 ago
Routing Descriptor Blocks:
  10.10.11.2 from 2.2.2.2, 00:05:09 ago, via GigabitEthernet0/0
    Route metric is 11, traffic share count is 1
  * 10.10.10.2, from 2.2.2.2, 00:05:09 ago, via GigabitEthernet0/3
    Route metric is 11, traffic share count is 1
Router#
```

```
[admin@MikroTik] > ip route print where ospf
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, R - rip, b - bgp, o - ospf, m - mme,
H - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS PREF-SRC GATEWAY DISTANCE
0 Ado 10.10.10.0/30 10.10.30.1 110
1 Ado 10.10.11.0/30 10.10.20.1 110
2 Ado 192.168.10.0/24 10.10.20.1 110
3 Ado 192.168.20.0/24 10.10.30.1 110
4 Ado 192.168.30.0/24 10.10.30.1 110
5 Ado 192.168.40.0/24 10.10.30.1 110
6 Ado 192.168.50.0/24 10.10.20.1 110
[admin@MikroTik] >
```

```
VPCS> ping 192.168.50.10
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=7.197 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=7.257 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=6.816 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=9.935 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=8.487 ms
VPCS>
```

```
VPCS> ping 192.168.50.10
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=7.234 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=5.390 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=6.306 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=6.030 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=5.855 ms
VPCS>
```

```
VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.40.1 4.823 ms 13.168 ms 3.707 ms
 2 10.10.10.2 7.138 ms 4.418 ms 4.430 ms
 3 *192.168.50.10 11.531 ms (ICMP type:3, code:3, Destination port unreachable)
```

```
VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.40.1 5.836 ms 3.170 ms 3.815 ms
 2 10.10.30.2 7.239 ms 3.662 ms 4.809 ms
 3 10.10.20.1 13.104 ms 6.234 ms 5.117 ms
 4 *192.168.50.10 8.631 ms (ICMP type:3, code:3, Destination port unreachable)
```

```
VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.40.1    5.164 ms  4.485 ms  3.393 ms
 2  10.10.10.2     6.618 ms  5.867 ms  4.695 ms
 3  *192.168.50.10  7.003 ms (ICMP type:3, code:3, Destination port unreachable)
```